

Heavy Quark Mass Formation through Hierarchical Vacuum Transformation

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Abstract

The Standard Model assigns an independent Yukawa coupling to each quark flavour, without any physical explanation for the observed mass hierarchy spanning five orders of magnitude. We propose that heavy quark masses arise as deformation energies stored in the vacuum itself: the up and down quarks are primary vacuum topologies, and the heavier quarks arise when injected energy compresses those topologies through successive levels of transformation.

Two primary vacuum topologies are considered: (i) an equilateral-triangle arrangement (up-type) and (ii) a collinear arrangement (down-type). Compression of either topology by the factor $x = r_0/r$ leads to the exact deformation-energy relation derived directly from the pair potential $V = -k/r + A/r^2$ without additional assumptions. The vacuum stiffness is considered as the local spring constant which scales as x^4 , making each successive compression level much harder than the one before. The proposed model identifies two hierarchical levels of vacuum transformation. We propose two hierarchical levels of vacuum transformation. The first-level vacuum transformation compresses each topology to its lowest Bohr level ($n = 1$), giving charm and, through a triangle-collinear topology transition, bottom. The second level vacuum transformation drives the up-type topology into the x^4 -stiffened nonlinear spring at level $n = 5$, giving top. A relativistic Bohr quantization condition together with the nonlinear vacuum stiffening yields the cubic Bohr ladder $x_n^2 = n^3 K$, where $K = \hbar c / (m_0 c^2 r_0)$. The mass of the observed charm, bottom, and top quark masses are reproduced with a maximum error 2.0% using no free parameters beyond the light-quark mass values. The strange quark stands outside both transformation levels as the ground state of the hadronic confinement potential. Its anomalously long lifetime follows directly from the absence of any lower confinement level accessible to the strong interaction, and the strangeness quantum number is identified as the AVF confinement quantum number.

This paper explains the masses of the three experimentally observed heavy quarks and addresses the anomalously long lifetime of the strange quark. The framework does not exclude the possibility of additional higher-order vacuum-transformation states under sufficiently extreme conditions. Whether such states become experimentally observable or manifest as new heavy quark resonances remains an open question.

1 Introduction

Heavy quarks are produced in sufficiently energetic particle collisions, such as those at the Large Hadron Collider (LHC) [1], through several processes including gluon–gluon fusion, quark–antiquark annihilation, and two-photon fusion [2]. Whenever the centre-of-mass energy exceeds the threshold $2m_Q$, a heavy quark–antiquark pair can be created. The six known quark flavours are organised into three generations. The first generation consists of the up (u) and down (d) quarks, the stable constituents of ordinary matter. The second generation contains the strange (s) and charm (c) quarks; the third contains the bottom (b) and top (t) quarks. Their masses and electric charges are listed in Table 1.

Table 1: Quark masses and electric charges [2].

Generation	Quark	Mass, charge	Quark	Mass, charge
First	up (u)	2.16 MeV, +2/3	down (d)	4.67 MeV, $-1/3$
Second	charm (c)	1,270 MeV, +2/3	strange (s)	93 MeV, $-1/3$
Third	top (t)	172,760 MeV, +2/3	bottom (b)	4,180 MeV, $-1/3$

Unlike first-generation quarks, the second- and third-generation quarks are unstable. They decay through weak interactions, producing lighter quarks together with leptons or additional hadrons. The characteristic lifetimes differ dramatically across the generations and are listed in Table 2.

Table 2: Observed weak decay transitions of the heavy quarks [2]. The strangeness lifetime is for the charged kaon K^+ ; charm and bottom lifetimes are for D^0 and B^0 mesons respectively.

Transition	CKM element	Lifetime	Character
$t \rightarrow b W^+$	$ V_{tb} \approx 0.999$	$\approx 5 \times 10^{-25}$ s	intra-generational
$c \rightarrow s W^+$	$ V_{cs} \approx 0.973$	$\approx 4 \times 10^{-13}$ s	intra-generational
$b \rightarrow c W^-$	$ V_{cb} \approx 0.041$	$\approx 1.5 \times 10^{-12}$ s	cross-generational
$s \rightarrow u W^-$	$ V_{us} \approx 0.224$	$\approx 1.2 \times 10^{-8}$ s	cross-generational

Three features of Table 2 are noteworthy. First, the top quark lifetime ($\sim 10^{-25}$ s) is shorter than the hadronisation timescale; the top decays as a free quark before forming a bound state. Second, the charm and bottom lifetimes (10^{-13} – 10^{-12} s) are set by weak transitions whose rates are governed by the Cabibbo-Kobayashi-Maskawa (CKM) matrix [2]. Third, the strange quark lifetime ($\sim 10^{-8}$ s) is anomalously long—ten trillion times longer than the strong-interaction timescale. This anomaly is one of the central questions of the present paper.

Within the Standard Model (SM), quark masses arise from Yukawa interactions with the Higgs field after electroweak symmetry breaking [3]:

$$m_q = \frac{y_q v}{\sqrt{2}}, \quad (1)$$

where $v \approx 246$ GeV is the Higgs vacuum expectation value and y_q is a Yukawa coupling fitted independently for each quark. Although Eq. (1) reproduces the observed masses, it does not explain any production mechanism. The six couplings y_q are free parameters

with no derivation from first principles. Similarly, the elements of the CKM matrix are phenomenological inputs rather than predictions. The SM does not address the following questions:

- i) Why do the heavy quarks exist? What physical mechanism generates states 10^2 – 10^5 times heavier than the up and down quarks?
- ii) Why does this specific mass hierarchy arise? What selects charm at 1.27 GeV, bottom at 4.18 GeV, and top at 172.8 GeV rather than some other set of values?
- iii) Why does the strange quark decay anomalously slowly, on a timescale ten trillion times longer than the strong-interaction scale?

The Active Vacuum Framework (AVF) [4] approaches the quark spectrum from a vacuum energy deformation perspective. In a recent paper [5], it is shown that only the up and down quarks are equilibrium configurations of the vacuum deformation field, considered here as primary vacuum topologies. Their masses arise from the balance between Coulomb attraction and short-range repulsion within the confined charge arrangement, without Yukawa couplings.

The core hypothesis of the present paper is that the four heavier quarks are not independent field species but quantised compression states of the two primary vacuum topologies. When a sufficiently energetic collision transfers energy into a confined quark topology, that energy is stored as vacuum deformation energy rather than radiated away. The deformed topology must settle at a quantised compression level determined by a relativistic Bohr condition in the vacuum’s own elastic medium. As compression increases, the vacuum stiffness grows as the fourth power of the compression factor, creating qualitatively distinct transformation levels. The physical narrative of the model follows the chain: Primary Vacuum Topologies $\rightarrow$ Vacuum Compression $\rightarrow$ x^4 Vacuum Stiffening $\rightarrow$ Hierarchical Vacuum Transformations $\rightarrow$ Heavy-Quark Mass Hierarchy.

We propose that heavy-quark masses arise through two discrete levels of vacuum transformation. Their mathematical structure and physical consequences are derived in subsequent Sections. The two levels and one special state are:

- i) **First Level Vacuum Transformation** ($n = 1$): the lowest Bohr level of each primary topology. The up-type triangle gives charm; a cross-topology transition gives bottom.
- ii) **Second Level Vacuum Transformation** ($n = 5$): the up-type topology compressed into the x^4 -stiffened nonlinear spring, giving top.
- iii) **Confinement ground state**: the strange quark, occupying the lowest rung of the hadronic confinement potential, below the first transformation level. It is not a vacuum-transformation state but a confinement state; its anomalously long lifetime follows directly from this classification.

The rest of the paper is organised as follows: Section 2 reviews the primary vacuum topologies and pair potentials, and introduces the concept of compression coordinate and deformation energy. Section 3 derives the two transformation levels and the Cubic Bohr Ladder. Section 4 works out the mass of each heavy quark individually. Section 5 compares predictions with experiment and identifies testable consequences.

One of the companion papers [?] presents the vacuum relaxation dynamics and the decay cascade $t \rightarrow b \rightarrow c \rightarrow s \rightarrow u, d$. The other companion paper [?] connects topology-transition amplitudes to the CKM matrix elements and measured lifetime hierarchy.

2 Primary Vacuum Topologies and Vacuum Compression

2.1 AVF pair potential

The AVF assigns a deformation potential to each pair of opposite-flavour charge quanta $(\xi_q, \bar{\xi}_q)$ at separation r [5]:

$$V(r) = -\frac{k}{r} + \frac{A}{r^2}, \quad (2)$$

where $k = e^2/(36\pi\epsilon_0) \approx 0.480 \text{ MeV fm}$ is determined by the primitive charge-quantum geometry, and $A > 0$ is a single free parameter fixed by the up-quark equilibrium mass. Same-flavour pairs interact only through the repulsive term $+k/r$; the attractive A/r^2 core is absent for same-flavour pairs.

The potential Eq. (2) has a minimum at $r_{\text{eq}} = 2A/k$ with well depth $-k^2/(4A)$. At separations much smaller than r_{eq} the A/r^2 core dominates, producing the strongly nonlinear stiffness under compression that is central to the mass hierarchy derived in §3.

2.2 Primary vacuum topologies

Two distinct multi-charge arrangements minimise the total interaction energy of the deformation field. These are the primary vacuum topologies; they represent the fundamental topological building blocks on which the entire heavy-quark mass hierarchy rests.

Up-type primary topology (equilateral triangle). One central charge quantum $\bar{\xi}_q$ with three ξ_q quanta at the vertices of an equilateral triangle of circumradius R . Summing all pair interactions:

$$U^{(u)}(R) = -\frac{(3 - \sqrt{3})k}{R} + \frac{3A}{R^2}. \quad (3)$$

Setting $dU^{(u)}/dR = 0$ gives the natural circumradius

$$r_0 = \frac{6A}{(3 - \sqrt{3})k}, \quad (4)$$

and requiring $-U^{(u)}(r_0) = m_u c^2$ fixes

$$A = \frac{k^2(2 - \sqrt{3})}{2m_u c^2} = 0.014286 \text{ MeV fm}^2, \quad (5)$$

yielding $r_0 = 0.1409 \text{ fm}$ with seed mass $m_u = 2.16 \text{ MeV}$. The three vertex charges orbit the centre tangentially and are the active outer charges governing the up-type quantisation.

Down-type primary topology (collinear). One central charge ξ_q and two outer charges $\bar{\xi}_q$ diametrically opposite at half-separation r :

$$U^{(d)}(r) = -\frac{3k}{2r} + \frac{2A}{r^2}. \quad (6)$$

The natural half-separation is $r_0^{(d)} = 8A/(3k) = 0.0794$ fm, with seed mass $m_d = 4.67$ MeV. The two outer charges oscillate radially along the collinear axis. Table 3 summarises the properties of the two primary vacuum topologies.

Table 3: Properties of the two primary vacuum topologies.

Topology	Geometry	Outer charges	Natural radius	Seed mass
Up-type	Equilateral triangle	$N = 3$ tangential	$r_0 = 0.1409$ fm	$m_u = 2.16$ MeV
Down-type	Collinear	$N = 2$ radial	$r_0^{(d)} = 0.0794$ fm	$m_d = 4.67$ MeV

2.3 Compression coordinate

When a primary vacuum topology is compressed by driving its active outer charges inward to radius $r < r_0$, the compression factor

$$x \equiv \frac{r_0}{r} > 1 \quad (7)$$

measures the degree of vacuum deformation. At $x = 1$ the topology is at its natural radius (seed mass); as x increases, the charges enter the repulsive A/r^2 core of the pair potential, storing progressively more deformation energy. The Lorentz factor of the outer charges at compression x is (see Appendix A)

$$\gamma(x) = (x - 1)^2 + 1 \approx x^2 \quad (x \gg 1). \quad (8)$$

For charm $\gamma \approx 588$; for top $\gamma \approx 8 \times 10^4$. Every heavy-quark compression level is deeply relativistic; a non-relativistic treatment is not applicable.

2.4 Vacuum deformation energy

Substituting $r = r_0/x$ into Eq. (3) and using Eq. (5) to eliminate A , the deformation energy above the natural-radius value is

$$E_{\text{def}}(x) = m_0 c^2 (x - 1)^2, \quad (9)$$

where $m_0 = m_u$ for the up-type topology and $m_0 = m_d$ for the down-type. The identical formula follows from Eq. (6) for the down-type by the same calculation. Equation (9) is exact; it involves no Taylor expansion or approximation.

The total mass of the compressed topology is

$$m(x) = m_0 [1 + (x - 1)^2]. \quad (10)$$

All four heavy quark masses and the light quarks at $x \approx 1$ are described by Eq. (10) with the appropriate m_0 and x . Given an observed mass m_q , the compression factor is

$$x_q = 1 + \sqrt{(m_q - m_0)/m_0}. \quad (11)$$

Table 4 lists the compression factors for the observed quark spectrum.

Table 4: Observed quark masses and derived compression factors. Base masses: $m_u = 2.16$ MeV, $m_d = 4.67$ MeV. Data from PDG 2024 [2].

Quark	Family	Transformation level	$m_q^{(\text{obs})}$ (MeV)	x_q
u, d	both	primary state ($x = 1$)	2.16, 4.67	≈ 1
s	down	confinement ground state	93	5.35
c	up	First Level, $n = 1$	1,270	25.23
b	down	First Level, $n = 1$ (cross)	4,180	30.90
t	up	Second Level, $n = 5$	172,760	283.8

3 Hierarchical Vacuum Transformations

We propose that the allowed compression levels of the primary vacuum topologies are quantised. The quantisation condition is a relativistic Bohr condition in the elastic medium of the vacuum, modified by the x^4 scaling of the vacuum stiffness. The two resulting levels, where the first level and the second level vacuum transformations account for the observed heavy-quark masses of the second and third generation, respectively.

3.1 First Level Vacuum Transformation

The first-level vacuum transformation is the lowest quantised compression of each primary vacuum topology. We derive the quantisation condition separately for the up-type (tangential) and down-type (cross-topology) cases.

3.1.1 Up-type: relativistic Bohr condition

For the up-type topology, the three vertex charge quantum orbits the centre tangentially (velocity perpendicular to the radial direction). Lorentz contraction, therefore, acts along the orbital arc and does not change the circumference $2\pi r_0$. The relativistic de Broglie wavelength at compression x is

$$\lambda(x) = \frac{2\pi\hbar}{m_u c x^2}, \quad (12)$$

where the x^2 denominator arises from the relativistic momentum $p = \gamma m_u v \approx m_u c x^2$ at large $\gamma \approx x^2$. The circular Bohr condition $n \lambda(x) = 2\pi r_0$ gives

$$x_n^2 = n K_{\text{up}}, \quad K_{\text{up}} = \frac{\hbar c}{m_u c^2 r_0} = 648.4. \quad (13)$$

At $n = 1$, the first-level vacuum transformation threshold for the up-type topology is

$$x_c \equiv \sqrt{K_{\text{up}}} = 25.46 \quad (\text{charm quark}). \quad (14)$$

3.1.2 Down-type: cross-topology transition

For the down-type topology the outer charges oscillate radially (velocity parallel to r). As shown in Appendix A, Lorentz contraction acts on both the cavity length and the de Broglie wavelength equally; the two factors cancel in the standing-wave condition and leave no x -dependent quantisation for the collinear topology in its own geometry. However, the two primary topologies transform into each other under sufficient compression:

- i) Compressing the equilateral triangle drives its three vertex charges toward a collinear arrangement.
- ii) Compressing the collinear arrangement drives its outer charges into a triangular arrangement.

At the first level vacuum transformation, the fully compressed collinear topology has transitioned into the triangular geometry. The natural orbit radius for this triangulated configuration is r_0 (the up-type natural radius), not $r_0^{(d)}$. The down-type outer charges (mass m_d) now orbit at r_0 , and the Bohr condition Eq. (13) applies with $m_u \rightarrow m_d$ and a charge-count factor $N_{\text{tri}}/N_{\text{col}} = 3/2$ (three active charges in the triangular arrangement versus two in the collinear):

$$K_{\text{cross}} = \frac{N_{\text{tri}}}{N_{\text{col}}} K_{\text{up}} = \frac{3}{2} \times 648.4 = 972.6. \quad (15)$$

The first level vacuum transformation threshold for the cross-topology case is

$$x_b \equiv \sqrt{K_{\text{cross}}} = 31.19 \quad (\text{bottom quark}). \quad (16)$$

3.2 Second Level Vacuum Transformation

The observed second-level vacuum transformation is the compression level assumed to occur when the x^4 -stiffened vacuum spring accommodates $n = 5$ Bohr quanta. IN the following, we show why the first level does not extend simply to the top.

Applying Eq. (13) to the observed top compression $x_t = 283.8$ gives

$$n_{\text{top}}^{(\text{linear})} = \frac{x_t^2}{K_{\text{up}}} = \frac{(283.8)^2}{648.4} \approx 124, \quad (17)$$

which is not a small integer. This signals that the spring constant has changed qualitatively between the charm and top scales. The local spring constant at compression x is

$$k_{\text{spring}}(x) = \left. \frac{d^2 U}{dr^2} \right|_{r=r_0/x} = \frac{6 m_u c^2 x^4}{r_0^2}, \quad (18)$$

which grows as x^4 . Between charm and top the spring has stiffened by

$$\frac{k_{\text{spring}}(x_t)}{k_{\text{spring}}(x_c)} = \left(\frac{x_t}{x_c} \right)^4 = (11.25)^4 \approx 16,000. \quad (19)$$

The top quark and the charm quark inhabit the same pair potential but qualitatively different, which corresponds to 16,000 times stiffer elastic regimes. The second-level vacuum transformation requires a Bohr condition that incorporates this x^4 scaling.

3.3 Quantised Compression and the Cubic Bohr Ladder

3.3.1 Derivation of the Cubic Bohr Ladder

At compression level n the outer charges satisfy Eq. (13) at leading order, giving $x_n \approx n^{1/2} x_1$. Substituting into Eq. (18):

$$k_{\text{spring}}(x_n) = \frac{6 m_u c^2 x_n^4}{r_0^2} \approx n^2 \times \frac{6 m_u c^2 x_1^4}{r_0^2} = n^2 k_{\text{spring}}(x_1). \quad (20)$$

The spring at level n is therefore n^2 -enhanced relative to the first level. A spring that is n^2 times stiffer has a natural oscillation frequency n times higher, so n Bohr quanta fit within the orbit at level n instead of one. The corrected Bohr condition is:

$$x_n^2 = \underbrace{n}_{\substack{\text{quanta in} \\ \text{stiffer spring}}} \times \underbrace{n^2 K_{\text{up}}}_{\substack{\text{spring-enhanced} \\ \text{Bohr parameter}}} = n^3 K_{\text{up}}. \quad (21)$$

Equation (21) is the cubic Bohr ladder. It can equivalently be written as

$$\frac{x_n}{x_c} = n^{3/2}, \quad (22)$$

so successive transformation levels scale as $n^{3/2}$ in the compression factor. For the consistency check, from Eq. (21), we can find $x_n = n^{3/2} x_1$. So $k_{\text{spring}}(x_n) = n^2 k_{\text{spring}}(x_1)$, which is exactly the n^2 -enhancement assumed in the derivation. The cubic Bohr ladder is, therefore, self-consistent.

3.3.2 Second Level Threshold

The second level vacuum transformation corresponds to $n = 5$ of the cubic Bohr ladder:

$$x_t \equiv \sqrt{5^3 K_{\text{up}}} = \sqrt{81,046} = 284.7 \quad (\text{top quark}). \quad (23)$$

The ratio $(x_t/x_c)^{2/3} = 11.25^{2/3} = 5.01$ confirms that top and charm are separated by the integer $n = 5$ on the cubic Bohr ladder. The same cubic Bohr ladder with K_{cross} applies to the cross-topology bottom:

$$x_b = \sqrt{1^3 \times K_{\text{cross}}} = \sqrt{972.6} = 31.19. \quad (24)$$

4 Heavy-Quark Mass Hierarchy

4.1 First Level Vacuum Transformation

Charm arises as $n = 1$ of the first level vacuum transformation for the up-type primary topology. Inserting $x_c = \sqrt{K_{\text{up}}} = 25.46$ into Eq. (10):

$$m_c^{(\text{pred})} = m_u [1 + (25.46 - 1)^2] = 1,295 \text{ MeV}, \quad m_c^{(\text{obs})} = 1,270 \text{ MeV}, \quad \text{error: } 2.0\%. \quad (25)$$

No free parameters beyond m_u and r_0 are used.

Bottom arises as $n = 1$ of the first level vacuum transformation for the down-type primary topology via the cross-topology transition. Inserting $x_b = \sqrt{K_{\text{cross}}} = 31.19$ into Eq. (10):

$$m_b^{(\text{pred})} = m_d [1 + (31.19 - 1)^2] = 4,261 \text{ MeV}, \quad (26)$$

This is within 1.9% error over the observed mass of the bottom quark: $m_b^{(\text{obs})} = 4,180 \text{ MeV}$. The ratio of charm and bottom thresholds follows from the charge-count ratio alone:

$$\frac{x_b}{x_c} = \sqrt{\frac{K_{\text{cross}}}{K_{\text{up}}}} = \sqrt{\frac{3}{2}} = 1.22474. \quad (27)$$

Using the observed $x_c = 25.23$ (rather than the predicted 25.46) gives $x_b = 25.23 \times 1.22474 = 30.900$, matching $x_b^{(\text{obs})} = 30.90$ to 0.001%. This agreement confirms that the topological factor $3/2$ is exact, and the 1.9% mass error arises from the 0.92% error in the predicted x_c .

4.2 Top Quark Mass

Top quark arises as $n = 5$ of the second level vacuum transformation. Inserting $x_t = \sqrt{5^3 K_{\text{up}}} = 284.7$ into Eq. (10):

$$m_t^{(\text{pred})} = m_u [1 + (284.7 - 1)^2] = 173,834 \text{ MeV}, \quad (28)$$

This is within 0.62% error over the observed top quark mass $m_t^{(\text{obs})} = 172,760 \text{ MeV}$. The exact integer $n = 5$ is verified by

$$n_{\text{top}} = \left(\frac{x_t^2}{K_{\text{up}}} \right)^{1/3} = \left(\frac{(283.8)^2}{648.4} \right)^{1/3} = 4.990 \approx 5 \quad (\text{one part in } 10^3). \quad (29)$$

Within the present framework, intermediate levels $n = 2, 3, 4$ on the cubic Bohr ladder corresponds to compressed states at $m \approx 11 \text{ GeV}$, 37 GeV , and 88 GeV respectively. These states are expected to possess extremely short lifetimes due to real- W emission, and are therefore unlikely to appear as long-lived hadronic resonances in current experiments. No particle corresponding to these predicted compression levels has yet been observed experimentally.

4.3 Strange Quark Mass

Inserting the observed mass $m_s^{(\text{obs})} = 93 \text{ MeV}$ into the universal mass formula (Eq. 11) gives

$$x_s = 1 + \sqrt{\frac{m_s - m_d}{m_d}} = 1 + \sqrt{\frac{93 - 4.67}{4.67}} = 1 + \sqrt{18.91} = 5.35. \quad (30)$$

Checking against the cubic Bohr ladder:

$$n_s = \left(\frac{x_s^2}{K_{\text{cross}}} \right)^{1/3} = \left(\frac{28.6}{972.6} \right)^{1/3} = 0.31. \quad (31)$$

Strange quark, therefore, does not sit on the cubic Bohr ladder. It is not a vacuum-transformation state; a different physical mechanism is required. Therefore, we have explored the WKB confinement condition for it.

4.3.1 WKB confinement condition.

The down-type collinear topology oscillates within a hadronic confinement bag of outer radius R_{conf} . In the ultra-relativistic limit the Bohr–Sommerfeld radial action is

$$J = \oint p_r dr \approx \frac{2 m_d c^2 (x_s^2 - 2x_s) R_{\text{conf}}}{\hbar c}. \quad (32)$$

Setting $J = \hbar$ (WKB ground state, $n_{\text{WKB}} = 1$) and substituting $x_s = 5.35$, $m_d = 4.67 \text{ MeV}$, $\hbar c = 197.33 \text{ MeV fm}$:

$$R_{\text{conf}} = \frac{\hbar c}{2 m_d c^2 (x_s^2 - 2x_s)} = \frac{197.33}{2 \times 4.67 \times (28.62 - 10.70)} = \frac{197.33}{167.5} = 1.178 \text{ fm}. \quad (33)$$

This equals $\Lambda_{\text{QCD}}^{-1}$ to within 2%, the natural hadronic confinement scale set by QCD and independently known from hadron spectroscopy. No parameter is fitted in this case, and x_s is fixed by the observed m_s , and R_{conf} is obtained as the output.

To confirm the mass formula closes consistently, we verify that $x_s = 5.35$ indeed yields $m_s = 93$ MeV:

$$m_s = m_d[1 + (x_s - 1)^2] = 4.67 \times [1 + (4.35)^2] = 4.67 \times 19.92 = 93.0 \text{ MeV}. \quad \checkmark \quad (34)$$

The agreement with the PDG central value (93_{-5}^{+11} MeV) is exact within the input precision. The physical identification is: the strange quark is the *WKB ground state* of the down-type collinear topology confined at the hadronic scale $\Lambda_{\text{QCD}}^{-1}$.

4.3.2 Long lifetime from absence of lower level.

The cubic Bohr ladder defines a decay cascade, such that a compressed topology at level n unwinds as n decreases, emitting a W boson at each step, on the strong-interaction timescale $\tau_{\text{strong}} \sim 10^{-23}$ s. The strange quark is at the lowest rung of the hadronic confinement potential, and there is no rung below it within the same topology family. The strong interaction has no mechanism to lower the confinement quantum number further; decay requires the weak interaction to change topology family (collinear $\rightarrow$ triangular, or vice versa), which operates on timescale $\sim 10^{-10}$ s.

The conserved quantum number S introduced phenomenologically by Gell-Mann and Nishijima [6, 7] is identified here as the AVF confinement quantum number: $|S| = 1$ when the collinear topology occupies its confinement ground state. Strong interactions conserve S because they act within a fixed topology family and cannot modify the confinement level. Weak interactions violate S because they change the topology family.

5 Discussion and Predictions

5.1 Comparison with experimental masses

Table 5 summarises all predictions of the present framework against PDG 2024 [2] values.

Table 5: Predicted and observed compression factors and masses for the presently observed heavy quarks. Inputs: $m_u = 2.16$ MeV, $m_d = 4.67$ MeV, $r_0 = 0.1409$ fm ($K_{\text{up}} = 648.4$, $K_{\text{cross}} = 972.6$) without any additional free parameters.

Quark	Level	n	$x^{(\text{pred})}$	$x^{(\text{obs})}$	$m^{(\text{pred})}$ (MeV)	$m^{(\text{obs})}$ (MeV)
charm	First Level	1	25.46	25.23	1,295	1,270
bottom	First Level (cross)	1	31.19	30.90	4,261	4,180
top	Second Level	5	284.7	283.8	173,834	172,760
strange	confinement GS	—	input	5.35	93	93

Table 6 classifies the presently observed quark spectrum by experimental status and vacuum state type within the framework.

Table 6: Classification of the experimentally observed quark spectrum.

Vacuum state	Quarks
Primary states (natural radius)	u, d
Observed transformed states	s (confinement GS), c, b, t
Possible higher transformed states	to be determined experimentally

5.2 Predictions of the framework

5.2.1 Intermediate vacuum transformation states.

The cubic Bohr ladder predicts additional quantised compression levels between the first and second vacuum transformations. Using Eq. (10), the intermediate levels $n = 2, 3, 4$ correspond to characteristic masses of approximately 11, 37, and 88 GeV respectively. The absence of experimentally established quarks at these masses suggests that such highly compressed vacuum configurations, if produced, relax extremely rapidly through electroweak processes before forming observable hadronic resonances. Their possible existence may nevertheless be reflected through broad resonant structures or anomalies in multi-boson production at future high-energy colliders.

5.2.2 Confinement radius.

The strange-quark confinement radius $R_{\text{conf}} = 1.178$ fm is a prediction of the present framework. It represents the natural size of the confinement ground state obtained from the WKB quantisation condition without introducing additional free parameters. Future lattice-QCD and strangeness-sensitive measurements may provide opportunities for testing this prediction.

5.2.3 Higher-order vacuum transformations.

The hierarchical vacuum-transformation mechanism is not restricted to the currently observed heavy-quark spectrum. In the Active Vacuum Framework, each heavy quark represents a quantised vacuum-compression state rather than a fundamental particle generation. Consequently, the framework does not impose an upper limit on the number of possible vacuum-transformation levels. Additional higher-order compressed vacuum states may exist at substantially larger compression factors and correspondingly higher masses. The absence of experimentally observed states beyond the top quark, however, does not mean that the hierarchy terminates there. Rather, it indicates that present collider energies have not yet accessed those vacuum-compression regimes, or that such highly compressed states relax too rapidly to produce experimentally distinguishable resonances. The present work explains the known heavy-quark spectrum without assuming that it represents the complete set of possible vacuum-transformation states.

5.3 Scope and limitations

The proposed heavy quark model establishes the mass-formation mechanism for the presently observed quark spectrum, but does not address three areas deferred to subsequent papers:

- i) *Decay dynamics*: the cascade $t \rightarrow b \rightarrow c \rightarrow s \rightarrow u, d$, transition rates, and qualitative lifetime hierarchy.
- ii) *CKM matrix and branching ratios*: topology-transition amplitudes and their identification with CKM matrix elements, including the CP-violating phase.
- iii) *Higher transformation levels*: whether additional vacuum-compression states exist at energies beyond current experimental reach is not determined by the present framework.

The four equilibrium inputs ($m_u, m_d, r_0, r_0^{(d)}$) are derived in the companion paper [5]. The integers $n = 1$ and $n = 5$ are not fitted parameters; they are the only values consistent with the observed quark masses. The confinement radius $R_{\text{conf}} = 1.178$ fm is a consequence of the WKB condition, and not a free parameter.

6 Summary and Conclusion

The hierarchical vacuum-transformation picture provides a unified interpretation of the presently observed heavy-quark mass formation through quantised vacuum-compression states. The presently known heavy quarks naturally organise into two hierarchical vacuum transformations together with the strange-quark confinement ground state. The framework itself does not require the vacuum-transformation hierarchy to terminate at the top quark; additional transformation levels may emerge under sufficiently energetic conditions beyond those currently accessible experimentally. The core of the physical narrative of the paper follows the chain: Primary Vacuum Topologies $\rightarrow$ Vacuum Compression $\rightarrow$ x^4 Vacuum Stiffening $\rightarrow$ Hierarchical Vacuum Transformations $\rightarrow$ Heavy-Quark Mass Hierarchy. The up and down quarks are primary vacuum states at their natural radii. When sufficient energy is injected, each primary topology is compressed to a quantised level set by the relativistic Bohr condition in the vacuum's elastic medium. The vacuum stiffness grows as x^4 , creating two qualitatively distinct transformation levels that account for the three observed heavy-quark masses. The strange quark stands apart as the ground state of the hadronic confinement potential. The main quantitative results established in this work are:

- i) The deformation-energy expression: $E_{\text{def}} = m_0 c^2 (x - 1)^2$, derived directly from the AVF pair potential.
- ii) Cubic vacuum-transformation ladder: $x_n^2 = n^3 K$, arising from the combination of relativistic Bohr quantisation and the x^4 nonlinear vacuum stiffening.
- iii) Topological charge-count relation: $x_b = x_c \sqrt{\frac{3}{2}}$, following directly from the ratio of active outer charges in the two primary vacuum topologies.
- iv) Prediction of the observed charm, bottom, and top quark masses with a maximum deviation of approximately 2%, using only the light-quark parameters.
- v) An open hierarchical vacuum-compression energy in which the currently observed heavy quarks represent the lowest experimentally accessible vacuum-transformation levels, while higher-order compressed states remain possibility without introducing additional assumptions.

A Lorentz Factor and Geometric Asymmetry

A.1 Lorentz Factor

At compression x the outer charges store kinetic energy $m_0 c^2 (x - 1)^2$, as follows from the mass formula $m(x) = m_0 [1 + (x - 1)^2]$ by subtracting the rest mass m_0 . Since the relativistic kinetic energy is $(\gamma - 1)m_0 c^2$, equating gives

$$\gamma - 1 = (x - 1)^2 \implies \gamma(x) = (x - 1)^2 + 1, \quad (35)$$

which is exact. For $x \gg 1$ this reduces to $\gamma \approx x^2$, used throughout the main text.

A.2 Geometric asymmetry between the two primary topologies

Up-type (tangential motion): In this case the outer charge moves perpendicular to the radius vector, so Lorentz contraction acts along the arc direction. Both the effective circumference and the de Broglie wavelength $\lambda = h/p$ are contracted by the same factor γ , so they cancel identically in the Bohr condition $n\lambda = 2\pi r_0$. The circumference $2\pi r_0$ thus enters the quantisation condition unchanged, and the Cubic Bohr Ladder (Eq. (13)) is well-defined.

Down-type (radial motion): In this case the outer charge moves parallel to r . At compression x the relativistic momentum is $p \approx \gamma m_0 c \approx x^2 m_0 c$ (using $\gamma \approx x^2$ from Eq. (35) and $v \approx c$), so the de Broglie wavelength scales as $\lambda_{\text{rel}} \propto x^{-2}$. The Lorentz-contracted half-separation is likewise $r_0^{(d)}/\gamma \propto r_0^{(d)}/x^2$. In the standing-wave condition $n \lambda_{\text{rel}}/2 = r_0^{(d)}/x^2$ both sides scale as x^{-2} , and the factors of x cancel completely. The quantum number n is therefore independent of x : no x -quantisation emerges from the collinear standing-wave approach.

The cross-topology quantisation of §3.1 resolves this impasse: the compressed collinear state transitions into the triangular arrangement, whose circular Bohr condition quantises x via Eq. (15), giving the Bottom branch of the cubic Bohr ladder.

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